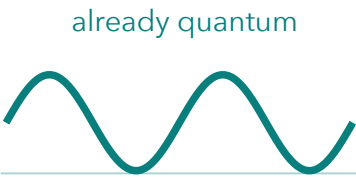


Exchange & correlation

DFT’s missing energy: same density, different electron-pair probabilities.

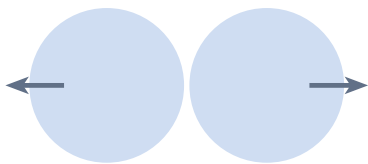
A **functional** maps the whole density to an energy. Kohn-Sham uses **noninteracting electrons at the same density**.



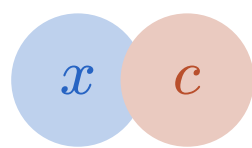
already quantum



nuclear attraction



smooth-charge repulsion



the missing correction

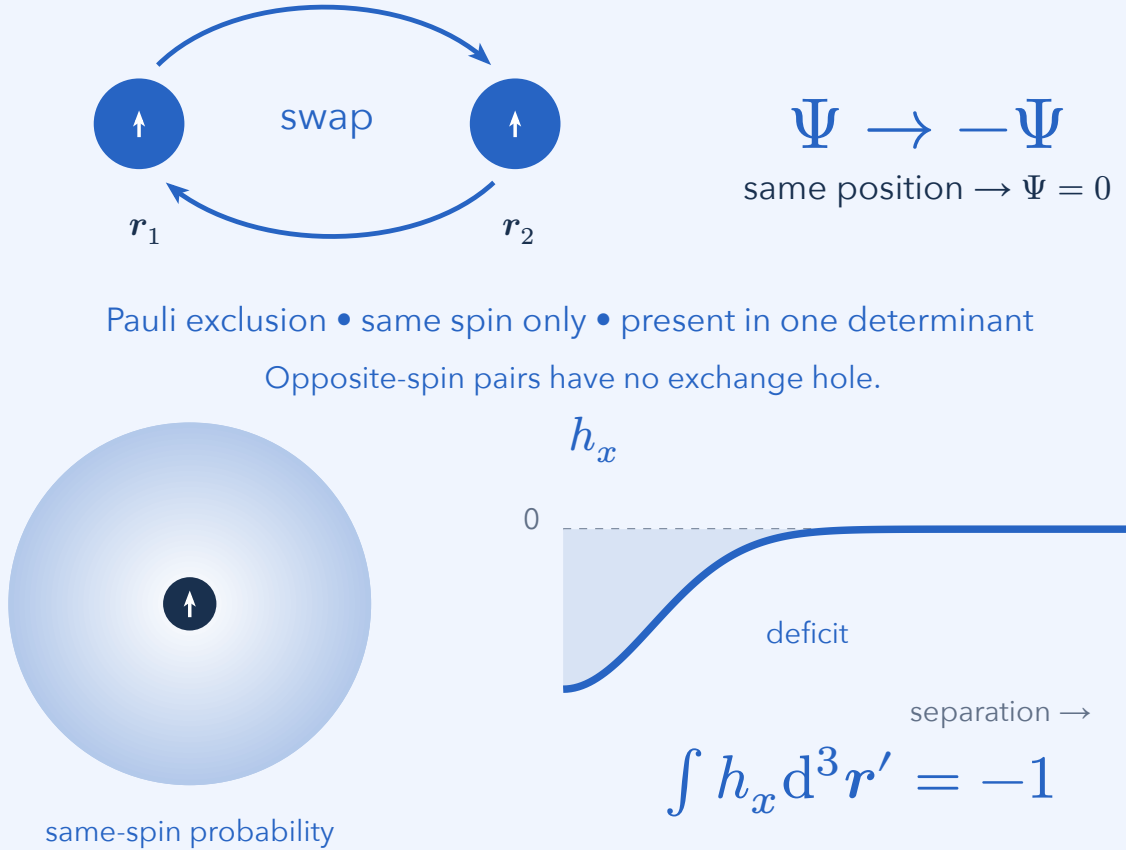
$$E[n] = T_s + V_{\text{ext}} + E_H + E_{\text{xc}}$$

KS kinetic energy

$E_{\text{xc}} = \underbrace{T - T_s}_{\text{kinetic correction}} + \underbrace{V_{ee} - E_H}_{\text{interaction correction}} = E_x + E_c$

EXCHANGE

Antisymmetry: swapping fermions changes the sign.



One missing electron. No extra force.

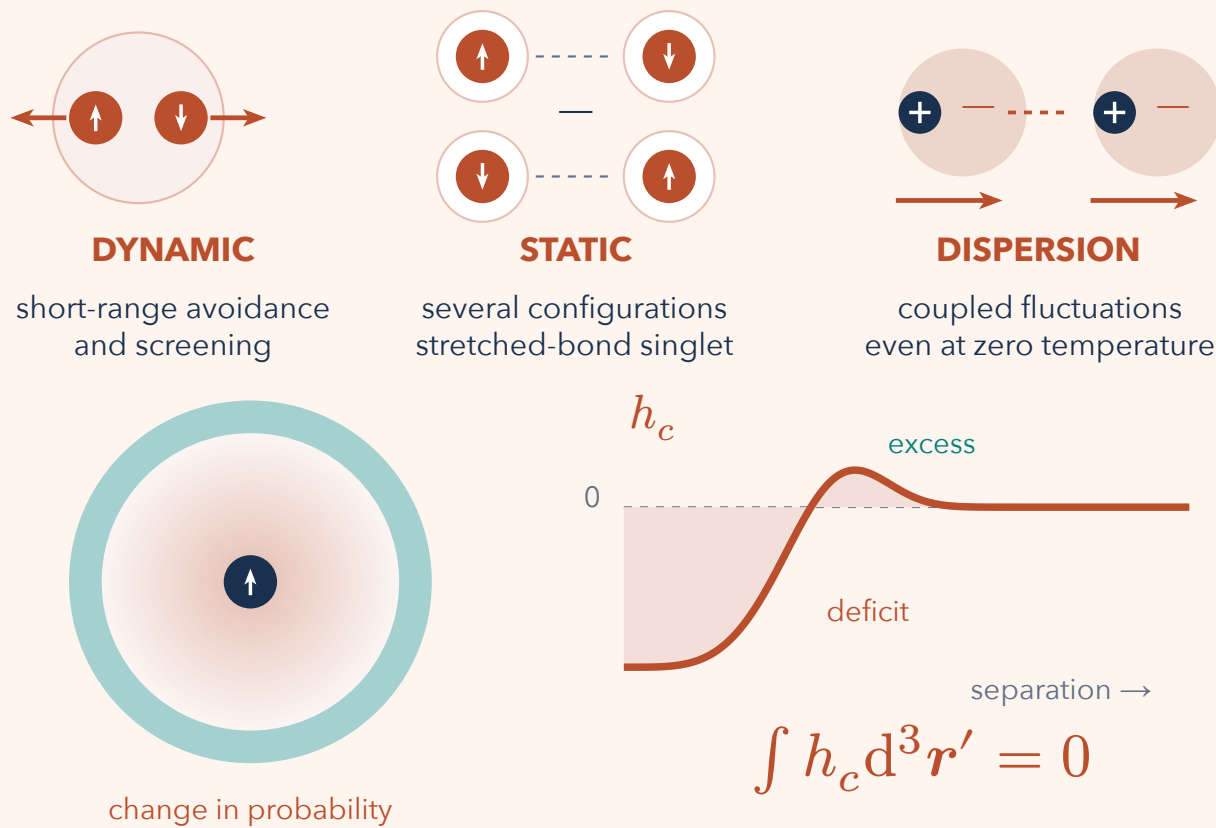
The hole persists without Coulomb repulsion.

$$E_x = \langle \Phi_s | \hat{V}_{ee} | \Phi_s \rangle - E_H \leq 0$$

Φ_s : antisymmetric KS orbital state. T , V_{ee} : exact interacting values at the same density.

CORRELATION

Joint probabilities beyond the KS determinant.



Same-spin and opposite-spin pairs.

A near deficit is balanced by an excess farther away.

$$E_c = \underbrace{T - T_s}_{T_c \geq 0} + \underbrace{V_{ee} - E_H - E_x}_{U_c} \leq 0$$

Correlation balances reduced repulsion against a kinetic cost.

ONE REFERENCE ELECTRON

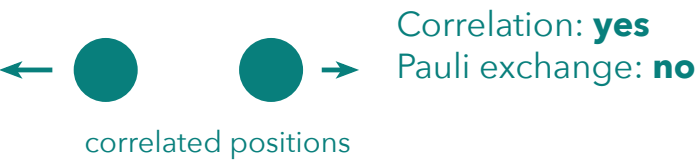
Only $N - 1$ others remain.

$$n_{\text{cond}} = n + h_x + h_c$$

$$\text{net hole: } -1 + 0 = -1$$

probability deficit, not an empty cavity

CLASSICAL CHARGES



ONE ELECTRON: NO SELF-REPUSSION


$$\begin{aligned} E_H + E_x &= 0 \\ E_c &= 0 \end{aligned}$$

Approximate XC can violate this cancellation.

XC FEEDS BACK INTO THE DENSITY

$$v_{\text{xc}} = \frac{\delta E_{\text{xc}}}{\delta n} \rightarrow \text{orbitals} \rightarrow n$$

Why approximate? Exact XC is formally defined, but no practical general expression is known.

LDA: local electron gas · GGA: density gradients · hybrids: orbital exchange. Ordinary LDA/GGA miss long-range dispersion.

Schematic maps / curves; map center = reference electron. Hole sums are 3D integrals. Fixed nuclei; spin labels suppressed; atomic units; nuclear repulsion added separately.